

Institute of Actuaries of India

Subject SA1– Health and Care

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Solution 1:

- i) - Lot of smaller schemes coming into the business mix, and, therefore, performance highly volatile due to some large claims [1]
- The claim reporting patterns changing from time to time leading to performance fluctuations caused by new groups into the mix/new TPA/ providers [1.5]
- Internal system issues/calculation errors [1]
- Unexpired risk reserve does not reflect the seasonality well enough due to extreme climatic changes, new types of groups/geographies coming into the business mix [1.5]
- IBNR methodology has not been adjusted to reflect the changing nature of reporting patterns [1]
- Outstanding to paid claims distorted by re-negotiations/resubmissions/reconciliations between providers/TPA and the Company [1]
- [5]
- ii) Reserving methods only change the timing of profit/loss arising and not the quantum of it as everything will have to come out in the wash eventually [1]
- AI alternative approach to IBNR is to Take a 'Book to Margin' approach that work as follows: [1]
- As on the date of the financials, for each group, derive expected Net Loss Ratio (NLR) (claims to net premium) blending [1]
- (i) the expected NLR at pricing [0.5]
- (ii) NLR based on actual experience emerging [0.5]
- ...in a such a manner the NLR starts with (i) in early duration and approaches to (ii) in later durations as time passes and the risk runs-off [1]
- .. the weights could be based on proportion of premium earned up to the date of the financials (proxy for how developed/credible experience is), and/or [0.5]
- ..size of premium earned up to the date of the financials (proxy for credibility) [0.5]
- Need to consider any regulatory considerations in adopting such an approach [0.5]
- ...and needs to test the impact on financials and smooth the impact over time if the methodology change itself leads to significant impact on the financials [1]
- [6]
- iii) QS is a proportional structure where the risk and premiums are split in the same proportion [0.5]
- XL is a non-proportional structure the reinsurer pays claims that exceed an attachment point and up to a limit ... [0.5]
- QS is preferable when the company is looking for
- Solvency capital relief [0.5]
- Technical support from the reinsurer [0.5]
- when expanding to new geographies/risks [0.5]
- to write more business without increasing capital [0.5]
- to go bit aggressive to taken on higher market share [0.5]

	A high level of QS reinsurance would imply the Company playing more of fronting the risk as opposed to taking the risk.	[0.5]
	There could be regulatory requirements on how much can be reinsured	[0.5]
	XL is preferable when the company is	
	- more experienced and now more confident taking the risk	[0.5]
	- concerned about passing on too much profit to the reinsurer under QS	[0.5]
	The choice will also be influenced by the availability and cost of reinsurance	[1]
		[5]
iv)	- Good idea as some will look for overseas coverage for better quality of care particularly for some severe illnesses and willing to pay for it	[1]
	- This will likely be a better deal for individuals compared to buying any individual products that cover overseas treatments	[1]
	- ... the individual products covering only overseas treatments might not be available	[0.5]
	- ... so, the top-up is a better fit for purpose	[0.5]
	- Premium collection can be through payroll reduction and so makes it convenient	[0.5]
	- ...and, likely make the product more persistent	[0.5]
	- If the competition is not offering such a top-up, this also helps make the basic plan sticky with the insurer	[0.5]
	- ...as individuals/corporates may not like to deal with two different insurers – one for basic and another for top-up	[0.5]
	- Those employers that offer more comprehensive (domestic + overseas) coverage may see this an opportunity to reduce the cost of insurance	[0.5]
	- Those employers that do not currently offer any health insurance might want to offer the basic plan if such plans do not exist currently	[0.5]
	- Additional administrative burden on the employer – so, some may not be willing to facilitate	[0.5]
	- ...could be complicated when there is low take-up for the top-up and additional underwriting involved	[0.5]
		[5]
v)	- The trick is to keep the cost lower to encourage higher take-up while at the same time benefits are seen to be 'good value for money'	[1]
	- Covers only specific critical illnesses or others for which local treatment options have been exhausted to keep the cost lower	[1]
	- Provide regional options in terms of geographical coverage as the cost of care varies significantly from one region to another (for example, treatment in the US are significantly heavier than that of India) to provide flexibility	[1]
	- Provide coverage for travel and companion at least partially to increase the marketing appeal of the product	[0.5]
	- Allow monthly mode of premium payment aligned to payroll to be convenient	[0.5]
	- Provide option to continue the top-up on an individual basis if the member leaves the group (eg. change of job), again to increase the marketing appeal	[0.5]

- ...include second opinion, counselling and concierge benefits as this is about severe illnesses and the individuals will require some level of handholding in those difficult times [1]
 - ...provide cash-less access to specific overseas network of providers to make it convenient for the members [0.5]
 - Minimal underwriting at inception of the plan through a medical declaration form [0.5]
 - Alternatively, use a combination of waiting period and leveraging claim experience under existing coverage to pre-underwrite – this will encourage higher take-up [1]
 - ...no underwriting at inception but with the requirement that the member was in the portfolio for at least 12 months [0.5]
 - ... new members may be allowed without underwriting as there is likely some level of medical selection when individuals take up employment [0.5]
 - ...the members should not have claims exceeding a certain threshold over past year [0.5]
 - ...the members should not have major medical illnesses in the past [0.5]
 - ...and/or PEC exclusion for a certain period [0.5]
 - ...if dependents are to be covered, the top-up should then be purchased for all dependents [0.5]
- [7]**
- vi) The Claims outgo is likely to comprise several components: Expected Take-up x Utilization of services x Average severity of cases x Case Mix Index. [1]
- Take up
- any similar experience or through a survey/consultation with the distribution/employers [1]
 - have regard to socio-eco profile of the target market and availability of close alternatives [1]
 - any surveys conducted by employee benefit consulting firms/others would help to assess 'what does the employees want and where the gap lies in the existing coverage' [0.5]
- Utilization
- Calibrated on similar profile of lives /product specifications/customer categorization and likely to be derived from in-house data, adjusted for the specific nature of this product [1]
 - If the company has a substantial and credible in-house data, it can carry out a GLM on various risk factors and tailor it to this top-up product (segment/network/overseas territory/nationality etc.) and use this as a base for projections [0.5]
 - Seek help from consultants and reinsurers that have data/experience on International private medical products [0.5]
 - Utilization will be influenced by the product design to a great extent. For instance, if product covers the cost of air travel, this will give an incentive for people to take it. [1]
- Average Severity of cases
- Requires more specific analysis as it is likely that the member may choose to go back to overseas country when treatment is likely not available locally. [0.5]

- Therefore, specific adjustment may be needed to reflect the higher severity and potentially, longer length of stay. [0.5]
- Here, using in-house data without adjustment is likely to be of little use. [0.5]
- Allowance need to be made on the overseas tariffs that are applicable- in particular, whether the overseas treatment shall be based on a network tariff or does the product provide reimbursement? [0.5]
- Mix of overseas country is likely to be a key issue – though it could be that the product covers only a certain number of countries. [0.5]
-this will need to be dynamic and linked to the nationality and also, territorial distance to a country which is able to dispense high quality services for these severe diseases. [0.5]
- Possibility of fraudulent activities may undermine the average severity – in particular, the LOS. [0.5]
- Possible currency devaluation may have large swings on the ultimate costs [0.5]
- If the company has a substantial and credible in-house data, it can carry out a GLM on various risk factors and tailor it to this top-end product (segment/network/overseas territory/nationality etc.) and use this as a base for projections [0.5]

Case Mix adjustment

- Case mix adjustment to reflect the fact that the conditions are likely to be major medical illnesses and likely at higher end of severity – such as cancers, organ transplant, Cardiovascular disease, Cerebrovascular disease etc. requiring lengthy stay in hospitals. [1]
- The information can be sourced from international health providers or health assistance companies. [0.5]
- The insurer may use in-house proprietary data/industry studies if there is enough and credible experience [0.5]

Rating factors to consider for premium structure - Age, gender, residence, occupation/industry, primary vs dependent and Geographical coverage [1]

Scenario testing by varying the assumptions that are more uncertain is important – present the scenarios to the senior leadership to ensure there is informed decision making [1]

[11]

- vii)** If pandemic is excluded, then the additional risk is limited and would be driven by [0.5]
- Govt/regulator forcing the insurers to pick up some or all of the costs [0.5]
 - Claims overlap – some part of pandemic claims may disguise as normal claims [0.5]
 - ...it may not be clear-cut what is covered or not in certain situations [0.5]
 - Reputation risk of declining pandemic claims especially when covered at least partly by some competitors [0.5]

If pandemic is covered, then risk of insolvency due to available capital being eaten up by extra claims and assets backing capital are worth less now than before [1]

Mispricing risk - for new business/renewals, uncertainties with respect to ascertaining the impact of the pandemic for pricing decision making...	[0.5]
- ...how long it will last? Pattern of infection and recovery?	[0.5]
- ...what sort of short term/long term morbidity/mortality implications?	[0.5]
- ...how wide-spread it will be and what extent the health care systems will cope with it?	[0.5]
- ...which lives (age groups/geography/gender/healthy/sub-standard risks etc.) does it affect what extent?	[0.5]
- ...how to interpret the past claims experience in terms of what extent it is under/overstated to feed into the experience rating	[0.5]
- ...how to interpret the past claims experience in terms of lag in claim reporting (lag might increase)	[0.5]
- ...how to interpret the past claims experience in terms of deferred treatments spilling over to different time periods	[0.5]
Risk of competition taking advantage of the situation to offer selective products with pandemic coverage to win tactical market share which your Board may not be supportive of	[1]
Operational risk associated with premium collection, ability to do medical underwriting etc.	[1]
Increased counter-party risks (eg. reinsurers/brokers defaulting on their payments)	[1]
	[9]

viii) Solvency ratio = available capital / required capital	[0.5]
Regulatory regimes define these and expect the solvency ratio to be above a certain threshold	[0.5]
Regulator may go easy on this requirement at extreme unprecedented times and may require insurers to put in place a specific action plan to build up the solvency ratio	[0.5]
The Required Capital may be a function of different approaches – based on the solvency regime in question.	[0.5]
For some jurisdiction, it may be factor based/risk based (such as RBC in the US and Canada and some Asian countries) or Solvency 2 based principle (risk-based principles or formula based)	[0.5]
The company would need to assess excess claims expected as a result of the pandemic using models that simulate the impact of such pandemics and estimate the recovery pattern	[1]
This may need to be reviewed frequently (say, monthly) depending upon the severity and expected duration of the pandemic as the impact might vary over time	[1]
Stress test the key assumptions underpinning its required capital model.	[0.5]
For instance, it can apply the following:	
- Investment returns fall of -20%	[0.5]
- Lapse increase of +20%	[0.5]
- Claims- Frequency and Severity increase of +20%	[0.5]

- Credit risk stress test of -20% on downgrades, premiums receivables (plus implications on possibility that the company does not receive the premiums in total), therefore, implying a high loss allowance in premiums which are uncollected. [1]

The company shall need to also allow for its risk mitigation strategies into the stress test. In particular, if it has invested in derivatives for hedging, it needs to allow the impact that this will have to offset the primary risk in the first place. [1]

The company shall also need to allow for residual or uncorrelated risks that may crop up when adopting risk mitigation strategies. In particular, there may be operational risks that would need to be allowed for. [1]

The company shall need to allow for correlation across all its business lines and geographical scope of coverage to allow for diversification and reduce potential required capital. [1]

Economic slowdown associated with the pandemic and its impact on the company's assets, investment returns and counterparties will have to be factored in – this will impact the available capital [1]

The company shall need to allow for any extra capital that it is raising from capital markets- either equity or high-quality debt instruments or contingent capital within the forecast stress model as this will increase the available capital. [1]

Choose an appropriate metric for the evaluation- for instance, use VaR99% or Conditional Tail Value at Risk 99% (which evaluates the quantum of the loss exceeding 99% upper limit within the scenarios). [1]

[9]

[57 Marks]

Solution 2:

- i) Considerations to sell it as a rider
 - a. The rider benefit could augment the benefits provided under the base product.
 - b. Hence, the rider premium would reflect the cost of the additional benefits. This could reduce the premium for the rider and make it more affordable.
 - c. Underwriting of rider could be linked to the underwriting of base product so no additional UW effort is needed.
 - d. Application process could be easy as most of the information, when selling to existing customers, is already available with the insurer
 - e. Mis-estimation risk is relatively low since the profitability of the rider could be evaluated in conjunction of base product. This is important since the incidence rates are not fully known.
 - f. Once the pandemic is over, it will be easier to stop selling the rider or renewing it while the base policy will continue. So the customer will remain with the company

Considerations to sell as standalone

- a. Easier to market as this will attract customer's attention.

- b. The benefit design, policy conditions and exclusions can be specifically designed for this product
- c. The application process can be simpler than other indemnity products with wider coverage
- d. Given limited coverage, premium is likely to be lower than indemnity which could increase the sales and may bring new customers.
- e. The insurer can then cross sell or up sell to new customers
- f. The product can also be designed as per regulatory requirements.
- g. There is a risk that it can be mis-sold, the customer may think the product covers other ailments as well. The messaging has to be clear to avoid any future customer complaint

[0.5 marks each, Max 6]

[6]

ii) a) Product design

- Benefit coverage
 - Are there any Fixed benefit payable and does it vary by SI? For e.g. daily fixed benefit during quarantine. How many days for which it is paid for?
 - Indemnity benefit during in-patient hospitalization- what additional costs will be covered other than hospitalization e.g. cost of protective gears etc
- What is excluded from the scope of cover? e.g. will benefit be payable only for institutional quarantine or for home quarantine as well.
- Boundary conditions –
 - min/max age – does the max entry age look appropriate? Are there regulatory guidelines around max entry age?
 - min/max Sum Insured- does the sum insured look appropriate? What could be the minimum SI that will provide sufficient coverage?
 - Can the term be less than 1 year? Can it be e.g. 3 months or 6 month or just 1 month. Given the short term nature of pandemic, should policy term of less than 1 year be offered? Does the regulation allow for less than one year term?
 - If the policy term is 1 year then should monthly premium mode be offered as the customer may lapse once the pandemic is over or once the individual contract COVID and recovers
 - Is there any waiting period for this product?
- Should the product only pay for first incidence or multiple incidences of COVID
- Individual floater or family floater- given that the infection could easily spread to other members of family, does it make sense to offer family floater? How to manage accumulation risk?
- Are there any specific exclusions that are over and above standard exclusions?
- Are similar products available in the market? If yes then how does the benefit offered under this product compared to others?

[1]

[1]

[2]

[0.5]

[1]

[0.5]

[1]

- Can the benefits offered be construed as too generous and thus leading to possible anti-selection?

[1]

[8]

b) Pricing & Reserving

- How the best estimates have been derived?
- What information / data has been used to justify the incidence rates and average claims cost? Do they look appropriate?
- How does the avg claims cost vary by network and non-network hospitals? How the average claims amount has been adjusted for additional claims cost due to COVID etc.
- What sensitivities have been carried out? What assumption is the profit very sensitive to?
- Are there scenarios regarding maximum probable loss. Is there sufficient capital available for the worst case scenario?
- Is there reinsurance available? If yes then what terms? If no then how will the additional volatility risk be managed
- Does the volume estimate look reasonable ?
- Are all pricing criteria being met ?
- Are there existing products in the market? Are the premiums comparable for a comparable benefits specifications?
- Is pricing appropriate for the underwriting of this product? If the product has minimal UW e.g. declaration of good health, then there are chances of people with co-morbid conditions taking the policy and it can increase the overall claims cost of COVID positive patients

[6]

Reserving

- What reserves will be held for this product and whether it will be sufficient?
- Given the product provide cover for pandemic,
- How to estimate the adequacy of IBNR? The past claims patterns mayn't necessary be relevant given there could be significant delays in claims notification due to lockdown or other restrictions.
- Given very low premium per policy, will there by additional reserves like premium deficiency reserves?

[2]

[8]

c) Underwriting and Claims

- What UW guidelines have been offered for this product? Are they less stringent than existing indemnity product
- Are the criteria comparable with other similar products in the market?
- Given that this product will cover infectious disease, what questionnaire is appropriate? E.g. short form or long form or simple declaration of good health. Should there be additional questions regarding past incidence of COVID?

- Are there question on life style diseases like diabetes and hypertension?
- Will pre-existing condition exclusion be applicable? If no then how should the additional risk due to chronic condition be underwritten.
- What are the claim triggers?
- Given that most of the claims will be for reimbursements, as a higher % of people may be under quarantine, does the claims team have sufficient staff to adjudicate the claims?
- How will the investigation be done specially given restricted movement during the lockdown period?

[1 mark each]

[8]

d) Sales and Marketing

- Which channels will be selling the product? Do they require additional training to sell the policy?
- What commission is paid and is it comparable with other product? Given that absolute premiums will be very low, the absolute commission amount could be very low. In such a scenario, will the agency channels be motivated enough to sell it?
- Given limited underwriting, simple product features and restricted movement during lockdown , can it be sold via online? The saving in commission can be passed back as lower premiums thereby increase the access and volume of business.
- How will the product be marketed and whether the cost is appropriate.
- What sales message be conveyed and whether it is clear and unambiguous.
- Does it meet regulatory requirements?
- Is there a Unique Selling Proposition for this product and is it being highlighted as part of the sales message?
- Will be product be sold across the countries or certain pockets where the incidence is high and hence, people will be more fearful?
- Which segment of population does the product intend to target and whether the channel and messaging is appropriate

[8]

- iii)**
- As the private hospitals are required to offer free treatment, they are likely to recover the cost from other patients who can afford treatment.
 - The individuals who will receive free treatment are likely to be uninsured so the insurer mayn't get any benefit of free treatment
 - The hospital is likely to increase charges for patients with insurance. This may increase the overall outgo of the insurers. Since the increased cost may not be priced in the premiums, then this may bring down insurer's profitability in the short term.
 - It could also increase the reserves and capital requirements for the insurer
 - Having said, if the insurer has already agreed tariff in advance with the hospitals / providers then the impact could be minimal or delayed. The hospital may renew the tariffs at a higher rates which could increase insurer's outgo and hence

[5]

premiums. If there is a lag between the renewal of tariffs and repricing exercise then the insurer's margins may get depressed for a short term

- There could be cases of abuse or overuse by provider which may need to be handled by the insurer.

[43 Marks]
